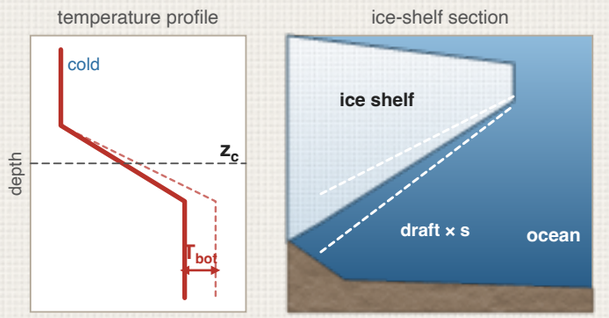
# Experiment 3: can a fast emulator replace the ocean model, and can it be trusted outside its training data?

A guide to the protocol: what is predicted from what (1), how it is tested (2), by which methods (3), and the questions fixed in advance (4).

## 1 What an emulator does

The 29 runs of Exp. 2 are the data: each pairs three numbers with one map of the basal melt rate.

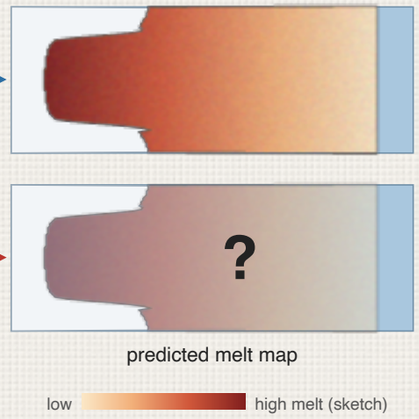



**Inputs: three numbers per run**

$T_{bot}$, deep-water temperature; $z_c$, thermocline depth; s, draft scale

**MITgcm ocean model**
solves the flow in the cavity
hours per run on 5 cores

**emulator (M0–M7)**
fitted on the training runs only
a fraction of a second per map



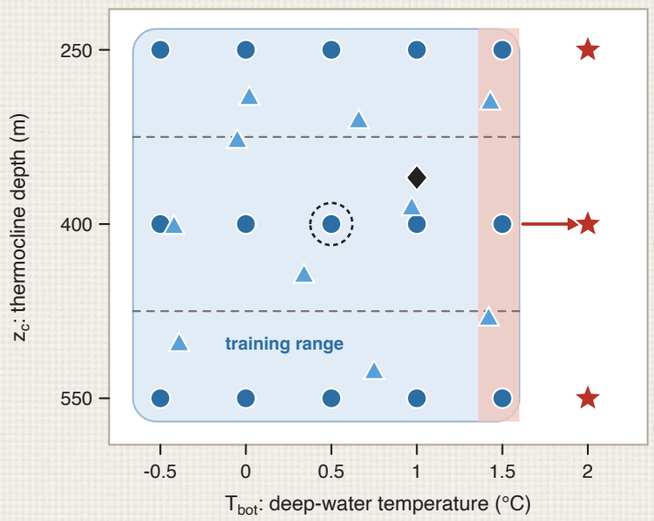

**Compared on runs the emulator has not seen:**

- error of the whole map (RMSE)
- error of the total melt of the shelf
- error near the grounding line, where the melt is strongest
- for M5–M7, also: is the stated uncertainty honest? (coverage, CRPS)

Metrics of Burgard et al. (2022), Eq. 32–35.

## 2 Where each method is tested

Each test hides some runs, fits on the rest and predicts the hidden ones.




15 runs on a grid, s = 1

10 runs with the draft scaled, s = 0.8 or 1.2

**Inside the training range**

**loro**
hide one run, predict it; repeat for each of the 25 runs.

**Outside it: extrapolation**

**zc_out**
hide the shallowest, middle or deepest third of $z_c$.

**warm_out**
fit on $T_{bot} \le 1.0$ °C, predict the 5 warmest runs.

**profile_test**
Ocean0, the ISOMIP+ standard run: a straight-line profile, no thermocline.

**final**
3 runs at $T_{bot} = 2.0$ °C, warmer than every training run; predicted once, with a frozen setup.

## 3 Eight methods, from pure physics to pure data

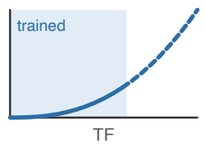
physics ⟷ data

| M1–M3 | M7 | M4 | M5 | M6 |
|---|---|---|---|---|
| melt ∝ TF or $TF^2$<br>one fitted constant | hybrid:<br>physics × network | regression per cell on $T_{bot}$, $z_c$, s | Gaussian process on map patterns | 5 neural networks<br>10 local inputs |

and a baseline, M0: the mean of the training maps, the score any method must beat.

**The idea behind the hybrid M7**

$$\text{melt} = K \cdot TF_+^{\,n} \times g_\theta(\text{geometry})$$




Physics sets how much: a power law in the local thermal forcing TF, with K and n fitted. It continues smoothly into warmer water.



A network sets where: it sees only the geometry (draft, distances, slope, s), never a temperature.

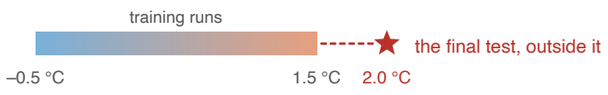
**So the network is never asked about a temperature it has not seen. M6, by contrast, takes temperature as an input and must extrapolate it.**

## 4 Three questions, fixed before any hidden run was predicted

Thresholds in notes/pd0_preregistro.md. Which method ranks first is reported, not hypothesised.

**H3a Does the physics help the hybrid extrapolate to warmer water?**

On the 3 final-test runs, M7 has a smaller error in the total melt than M6, on average and in at least 2 of the 3 runs.



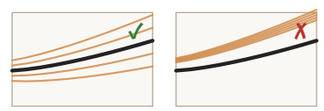

**H3b Does the hybrid lose little inside the training range?**

In loro, the map error (RMSE) of M7 is at most 1.10 times that of M6.

The price of the physics constraint:
the hybrid may fit the training range a bit worse.

**H3c Does the network ensemble know when it is wrong?**

In the 33 extrapolation cases, never a spread below 5 % together with an error above 20 % (a replication of Ockenden et al., 2026).




black: model; orange: 5 members

Not read: differences below the noise floor of the runs themselves (0.157 m $a^{-1}$ for the map, 1.3 % for the total melt).

A schematic: no result of Experiment 3 is drawn. The melt maps, curves and ensemble members are sketches; the ice-shelf outline is the ISOMIP+ grounding line.